

Article

Geographic Named Entity Matching and Evaluation Recommendation Using Multi-Objective Tasks: A Study Integrating a Large Language Model (LLM) and Retrieval-Augmented Generation (RAG)

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Abstract: Geographical named entity matching, a crucial step in address encoding, aims to enhance address resolution accuracy through the precise identification and linkage of geographical named entity data. However, existing approaches tend to ignore the spatial information of entities, leading to misclassification. Drawing on the human process of searching for addresses, this study proposes a multi-objective learning model named GNEMM that integrates the semantic and spatial information of geographical named entities. To further mimic the human cognitive process during address search, it incorporates the Retrieval-Augmented Generation (RAG) technique. By integrating newly added external address data with an advanced large language model (LLM) like GPT-4, it achieves precise address evaluation and recommendation. The model was tested using a standard geographical named entity dataset from Shandong Province, focusing on three sub-tasks: element segmentation, matching, and spatial similarity score prediction. The experimental results indicate that the method achieves a geographical named entity matching accuracy of up to 99%, with improvements of 10% and 5% in the segmentation and prediction sub-tasks. GNEMM performs best in address-matching tasks of various scales, and the vectors extracted by GNEMM perform best in the downstream retrieval and matching of various address types, which verifies its applicability in geographical named entity recommendation applications.

Keywords: geographical named entity matching; multi-objective; Retrieval-Augmented Generation (RAG); large language model (LLM); geographical named entity recommendation



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1. Introduction

In today's rapidly digitalizing and networked world, Geographic Information Systems (GISs) play an indispensable role in our daily lives and emergency responses. Particularly in managing natural disasters such as earthquakes, floods and fires, the ability to accurately and swiftly match and identify geographic named entities directly impacts the effectiveness of emergency measures. For instance, on social media platforms, real-time messages related to geographic locations [1] posted by users must be precisely parsed and applied, placing higher demands on geographic named entity matching systems.

Geographic named entity matching is a highly challenging natural language processing task [2–8], primarily due to the diversity and dynamic nature of geographic named entities, where traditional methods often struggle to adapt. The foundation of geographic named entity matching lies in understanding the semantics and spatial information of geographic entities and comparing them with existing geographic records in databases to identify and link entities with similar geographic meanings. Traditional methods of geographic named entity matching, such as those based on character similarity and elements, although effective in some scenarios, often fail to meet the demands for high precision and efficiency in complex cases.

Traditional rule-based address matching methods can be categorized into two types, one based on character similarity, distinguishing similar address elements through character-by-character comparison [9–11], and another based on address components, where addresses are parsed into individual components and matches are evaluated based on hierarchical structures and matching rates [12–15]. While these methods perform well with standardized or specific address formats, they typically require significant human resources and time investment. Currently, methods based on deep neural networks [16–18] are employed for geographic named matching, which standardizes and precisely matches addresses by processing address data and training models, thereby enhancing the understanding of address semantic information. These deep neural-network-based methods not only offer greater flexibility, particularly in handling non-standard addresses, but also enhance generality.

However, as demonstrated in the case from Table 1, even though geographic named entities may share the same latitude and longitude coordinates, their meanings can differ. For instance, “27 Shanda South Road, Licheng District, Jinan, Shandong Province” and “Shandong University (Central Campus)” are geographically co-located, yet the former is a specific address point, whereas the latter refers to a broader campus area. This situation illustrates that if a system overly relies on spatial proximity while ignoring the semantic differences in place names, geographic named entities that should match might be incorrectly marked as mismatches.

Table 1. Samples of geographical named entity pair.

No	Geographic Named Entity Pair	Coordinates	Match or Not
1	Building 5, Ginza Center, Harmony Square, 16th Floor, 1611, Jingshi Road 22799	(116.972, 36.651)	True
	Room 611, 6th Floor, Building 5, Ginza Center, Harmony Square, Jingshi Road 22799		
2	27 Shanda South Road, Licheng District, Jinan City, Shandong Province Shandong University (Central Campus)	(117.054, 36.675)	False

To effectively address the challenges of geographic named entity matching, this study introduces a multi-objective Geographical Named Entity Matching Model (GNEMM) that operates independently of a standard geographic named entity reference database. The model relies solely on the semantic information in the text and corresponding geographic spatial information to solve matching issues between entities with identical geographic locations.

While the Geographical Named Entity Matching Model (GNEMM) generally performs well, it struggles with entities that have similar spatial positions but different descriptions, particularly those that refer to the same location but have different names. In recent years, large language models (LLMs) like OpenAI’s GPT series have demonstrated exceptional capabilities in understanding and generating human language. These models, pre-trained on vast datasets, have learned the deep semantics and structural rules of language, enabling excellent performance across various natural language processing tasks. To address

these challenges, this study integrates Retrieval-Augmented Generation (RAG) technology, which combines traditional generative models with advanced retrieval mechanisms to fetch information related to the query location from extensive external knowledge bases. This information is then synthesized with model prompts for comprehensive analysis. Innovatively, we merge the RAG technique with the GNEMM's matching results, blending their strengths to produce a model that outputs the geographical named entities most relevant and semantically close to user queries, accomplishing a RAG-based recommendation task for geographical named entities. This approach not only enhances the precision of matching but also enriches the model's deep semantic understanding of geographical named entities.

The main contributions of this study are summarized as follows:

- (1) This study innovatively introduces a multi-objective Geographical Named Entity Matching Model (GNEMM), dedicated to addressing three sub-tasks: segmentation of geographical named entity features, matching of these entities, and prediction of spatial similarity scores. The model integrates semantic information and spatial data of geographical named entities, providing a comprehensive solution for their analysis.
- (2) By dynamically optimizing the weighting of each sub-task, optimal synergy between tasks was achieved, significantly enhancing the precision of geographical named entity matching. A series of detailed experiments demonstrated the model's superior accuracy in handling complex geographical named entity data, effectively validating the model's excellence.
- (3) The study applies Retrieval-Augmented Generation (RAG) technology, based on large language models (LLMs), to the task of recommending geographical named entities. By leveraging the robust generative capabilities of LLMs combined with extensive information from external databases, the model significantly deepens its understanding and processing of geographical texts. Particularly in matching and recommending geographically named entities closely related to query semantics, this technique enhances both relevance and accuracy.

The remainder of this paper is divided into four sections. Section 2 reviews related work in geographic named entity matching, geographic text recommendation, and Retrieval-Augmented Generation (RAG). Section 3 details the methodology and experimental data used in this study. The findings and their implications are discussed in Section 4, where we present our experimental results. Section 5 summarizes the research findings and discusses future work.

2. Related Work

In this section on related work, we review the evolution in the field of geographic named entity matching, highlighting the shift from traditional machine learning techniques to advanced deep neural network approaches. Furthermore, we examine the latest developments in geographic named entity matching in contemporary applications, particularly the use of Retrieval-Augmented Generation (RAG) technology in geographic text recommendation and entity matching.

2.1. Geographical Named Entity Matching

Methods based on deep neural network language models represent the latest approach for geographic named entity matching [2,19]. These methods aim to address the challenge posed by the vast amounts of non-standardized data with standardized location tags generated by the internet's rapid growth. Early researchers employed classical machine learning techniques and deep neural networks to extract semantic features from geographic named entity texts for matching purposes. For instance, Acheson et al. used a rule-based random forest method for cross-gazetteer matching [20], while Comber [21] employed CRF and

word2vec for address matching, enabling semantic extraction without the need for complex rules. However, these methods could only capture shallow semantic features. With the advancement of deep learning, more researchers have applied deep learning methods to geographic named entity matching tasks. Zhang et al. proposed a convolutional neural network (W-TextCNN) for Chinese address pattern classification [22]. The popularity of gated mechanisms in recurrent neural networks has led to numerous studies on geographic named entity matching and standardization using LSTM and GRU [23–26]. Shan et al. enriched the semantic information of addresses by searching and collecting address-related data from the internet as a corpus [27]. They used word2vec to obtain feature vectors of addresses and trained an address representation model combining attention mechanisms and LSTM to extract address vectors. Santos et al. utilized a deep neural network based on a bidirectional GRU for toponym matching [16]. Li et al. introduced the hierarchical relationship between address elements into their model, proposing a BiLSTM-based multi-task learning method [17]. Subsequently, the self-attention mechanism was introduced into Geographical Named Entity Matching Models. Lin et al. [28] employed the classical Enhanced Sequence Inference Model (ESIM) to model address record pairs [29]. With the widespread use of pre-trained language models, Qian et al. combined BERT and LSTM to propose a region-based hierarchical Chinese address geolocation method [30]. Additionally, Xu et al. developed a BERT-based model for extracting address semantic representations, achieving the integration of address semantics and geospatial information [18]. However, this method only performs simple concatenation of semantic and spatial features, failing to achieve the deep integration of geographic named entities' spatial and semantic information. Currently, most existing methods tend to overlook the spatial information of entities, focusing primarily on semantic features [31,32]. In summary, the most advanced geographic named entity matching methods currently rely on deep learning models to learn the semantic features of geographic names from text and measure the semantic similarity between geographic named entities. While these methods achieve high precision in matching, they have certain limitations. To address these limitations, this study proposes a multi-objective matching method that integrates the spatial and semantic information of geographic named entities.

2.2. Geographical Text Recommendation Based on Retrieval-Augmented Generation

In the domain of geographic information science, which deals with the acquisition, storage, processing, and visualization of geographic data, the proliferation of location-based social network services [33–36] has drawn increasing attention to geographic text recommendation tasks [36,37].

Currently, there are three main approaches to geographic text recommendation: content-based methods, collaborative filtering methods, and deep-learning-based methods. Content-based recommendation methods analyze the content of geographic texts, such as tags, categories, and descriptions, to calculate the similarity between the query text and standard texts, and then make recommendations [38]. These methods focus primarily on the content and features of geographic texts, requiring the appropriate representation and extraction of these contents. Common models include TF-IDF, LDA, and word2vec [39,40]. Collaborative filtering methods can be divided into two categories: user-based and item-based. User-based collaborative filtering analyzes the historical behavior data of users who query geographic texts, identifies a group of users with similar behavior to the target user, calculates the similarity between these users and the geographic text being queried, and recommends related geographic texts based on the most similar users. Item-based collaborative filtering, on the other hand, calculates the similarity between geographic texts and recommends the most similar geographic texts based on the target user's previous preferences [41,42]. Deep-learning-based recommendation methods build deep neural

network models to learn user preferences from vast amounts of geographic location data, model and analyze users' historical interests, and provide personalized geographic text recommendations. Deep-learning-based recommendation methods consist of two stages: feature extraction and recommendation prediction. In the feature extraction stage, algorithms use deep neural networks to extract relevant features from historical location information and geographic text attributes. In the recommendation prediction stage, algorithms use these features to predict users' interest in a specific geographic text and recommend texts that match users' interests [43]. Common deep learning models include LSTM and GRU [44,45]. With the development of large language model [46–48] technology, the Retrieval-Augmented Generation (RAG) framework was proposed initially by Patrick Lewis et al. [49] to address the limitations of large language models in handling domain-specific issues. Gao et al. reintroduced the RAG approach to mitigate the 'hallucination' phenomenon of large language models. Various RAG styles have since been proposed, generally categorized into Naive RAG, Advanced RAG, and Modular RAG, each focusing on different retrieval, generation, and enhancement techniques [50]. This framework enhances prediction capabilities by retrieving relevant information from a vast number of documents during the question-answering or text-generation stage and incorporating this information into the response. For example, Lin et al. [51] used the Retrieval-Augmented Dual Instruction Tuning (RA-DIT) method to enable pre-trained large language models with retrieval capabilities, significantly improving their performance in knowledge-intensive tasks. Wang-Chiew Tan et al. [52] innovated the RAG model architecture to improve the efficiency and accuracy of query answering tasks, while Lin et al. [53] enhanced retrieval performance and answer quality in visual question-answering tasks through fine-grained multimodal retrieval. Inspired by these developments, we propose an RAG-based address recommendation framework that uses new addresses and related information as an external knowledge base, combined with matching addresses from Geographical Named Entity Matching Models, to conduct address recommendation tasks.

3. Research Data and Methodology

In this study, the model training dataset, derived from a standard government-collected dataset, underwent basic preprocessing to construct a dataset for multiple objectives, with the training, validation, and test sets distributed in proportions of 70%, 20%, and 10%, respectively. This dataset division method aligns with the findings of Lin et al. [28], who demonstrated that similar proportions provide an optimal balance for training deep learning models while ensuring adequate validation and independent test evaluations across various domains. This dataset, rich in geographical named entities and precise spatial coordinates, was chosen for its extensive coverage and detailed granularity, making it ideally suited for geographical named entity matching tasks. This split ensures that the model has sufficient data for training while preserving enough data for robust validation and testing, thus allowing for effective evaluation of the model's generalization capability. Initially, referencing the architecture of Transformers, we utilized the BERT [54–56] framework as the encoder, removed BERT's final output module, and incorporated multiple sub-task modules as decoders to build the Geographical Named Entity Matching Model (GNEMM). Specifically, the sub-task modules are designed to operate in conjunction with one another, forming a cohesive multi-task framework that enhances the robustness and accuracy of the model. These sub-tasks include Geographical Named Entity Element Segmentation (corresponding to Section 3.2), Geographical Named Entity Element Matching (corresponding to Section 3.3), and Geographical Named Entity Spatial Similarity Score Prediction (corresponding to Section 3.4). Subsequently, we conducted experiments on sub-task weight combinations, sub-task combinations, and comparative experiments on

GNEMM. We then introduced the Retrieval-Augmented Generation (RAG) approach to undertake address recommendation tasks, initially verifying the effectiveness of our constructed model. In the RAG tasks, we combined geographical named entities matched by the GNEMM model with external geographical named entities information retrieved through RAG, employed the Chain of Thought (COT) method for enhancement, and ultimately inputted into the large language model gpt-4-turbo to complete the final address recommendation task. The overall research route is shown in Figure 1.

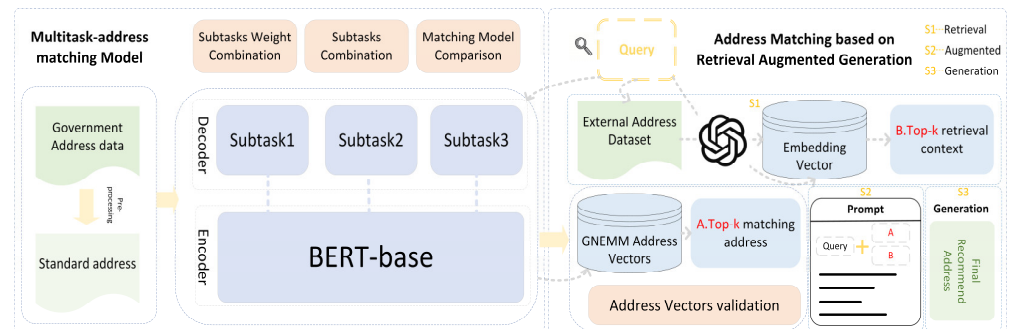


Figure 1. Research overall route.

3.1. Subsection

The experimental data include a government-collected address dataset, which is quite standard and, after simple preprocessing, is used for training models to deeply understand the semantic features of geographic named entities. The preprocessing of geographic named entity data employs the Markbert [57] strategy for segmenting geographic named entity texts, using the special marker [SEP] as a delimiter and inserting the [CLS] identifier at the beginning of the input text sequence as a sentence-initial marker. We use the “jieba” Chinese word segmentation library in Python (version 3.7) in its precise mode for segmenting geographic named entity texts. To prevent the jieba library from over-segmenting geographic named entity elements, we additionally loaded multiple address dictionaries from Jinan, Shandong Province, as custom dictionaries. This enables the jieba library to segment geographical named entity elements more accurately based on the address information in the custom dictionaries. For example, by using “Lixia District, Jinan City, Shandong Province”, we load the Jieba Chinese dictionary, which includes major cities and geographical names in Shandong. This ensures the accurate segmentation of geographical entities such as “Shandong”, “Jinan”, and “Lixia” during the text processing. The toponym dictionaries used are from publicly available sources such as the Sogou Pinyin dictionary, Tsinghua University’s Open Chinese Lexicon (THUOCL), and toponym databases provided by government agencies. Specific segmentation examples are shown in Figure 2, and the results of other segmentation instances are shown in Table 2.

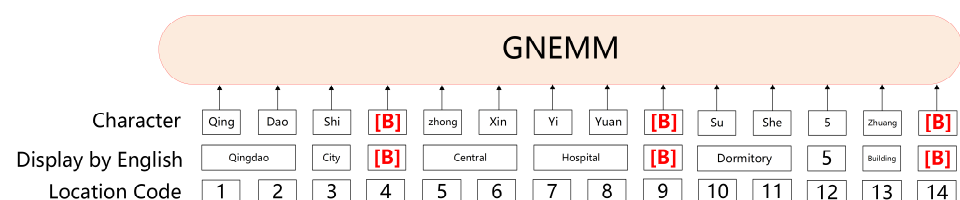


Figure 2. Geographic named entity segmentation example.

We designed a matching dataset integrating three subtask objectives, as shown in Table 3. We segmented based on the structure of the created matching dataset and compiled statistics on the dataset information, as shown in Table 4.

Table 2. Example of segmented geographical named entity text.

Geogapchical Named Entity	Segmentation Results
AoShengDaSha 1 HaoLou 22 Ceng	Aosheng Building/Building 1/22nd floor
JiNanLu 169 ZhengXi FangXiang 100 Mi	Jinan Road /169/ due west direction/100 m
ShanDongSheng JiNanShi HuaiyinQu DuanBeiDongLu 13 Hao	Shandong Province/Jinan City/Huaiyin District/Duan Beigong Road/No. 13

Table 3. Datasets structure of Geographic Named Entity matching.

Field Name	Field Definition	Field Example
geographic_entity_pair	Geographic entity text pair, using the special symbol [SEP] as a delimiter	Quancheng Square [SEP] No. 99 Luoyuan Street, Lixia District, Jinan City
element_segment_label	Geographic entity element segmentation label, using 0 and 1 alternately to indicate elements in the text, and 2 to indicate the special symbol [SEP]	[0 0 0 0 2 2 2 2 2 2 0 0 0 1 1 1 0 0 0 0 1 1 1]
match_label	Indicates whether the two geographic entities match, 0 for mismatch, 1 for match	1
similarity_score_label	Indicates the spatial similarity score between the two geographic entities, ranging from [0,5], with larger values indicating closer spatial proximity between the two geographic entities	5.0

Table 4. Statistical information of the spatial–textual matching dataset for geographical named entities.

Statistical Feature	Statistical Value
Total number of labeled geographic entity pairs	2,267,488
Number of matched geographic entity pairs	694,812
Number of unmatched geographic entity pairs	1,572,676
Average length difference of all labeled geographic entity pairs (in terms of characters)	2.43
Average length difference of matched geographic entity pairs	2.02
Average length difference of unmatched geographic entity pairs	2.62
Average Jaccard similarity coefficient of all labeled geographic entity pairs	0.48
Average Jaccard similarity coefficient of matched geographic entity pairs	0.75
Average Jaccard similarity coefficient of unmatched geographic entity pairs	0.36
Number of geographic entity pairs with spatial similarity score in the range [0, 1]	1358
Number of geographic entity pairs with spatial similarity score in the range [1, 2]	17,511
Number of geographic entity pairs with spatial similarity score in the range [2, 3]	75,598
Number of geographic entity pairs with spatial similarity score in the range [3, 4]	181,280

After segmenting geographic named entities using the “jieba” library and custom dictionaries, we employed a Chinese BERT model to deeply mine the semantic information of geographic named entity texts. This model utilizes multiple stacked self-attention modules, designed to enhance the extraction of semantic features significantly. To optimize performance, the model incorporates residual connections, layer normalization, and a fully connected feed-forward network, adhering to the Transformer architecture proposed by the Google research team. For this study, the decoder component of the traditional Transformer was omitted to better suit the specifics of our geographic named entity corpus and the associated pre-training tasks. Each multi-head self-attention module in GNEMM is defined as a Geographical Named Entity Transformer Module, designed to refine the understanding of complex semantic relationships within geographic texts.

From Table 4, it is evident that our previous processing methods were effective in constructing non-matching pairs of geographic named entities, meaning the non-matching

pairs were not completely dissimilar in their text descriptions. The Jaccard similarity coefficient shows greater differences in text descriptions between non-matching pairs compared to matching ones. Most pairs' spatial similarity scores are concentrated in the range. To reduce the cost of manual annotation, we grouped geographic named entity data by spatial coordinates, automatically constructed matching pairs, and assigned them a spatial similarity score of 5. While traversing the remaining data, we built non-matching pairs with the geographic named entities in the matching pairs, calculating corresponding spatial similarity scores based on spatial coordinate information. Additionally, to enhance data diversity and prevent repetition, we tracked the usage frequency of each geographic named entity datum, discontinuing use after five occurrences. When constructing non-matching pairs, we calculated the Jaccard similarity between the texts of the two entities, if the similarity score was 0, there was a 50% chance that the geographic named entity text would not be used.

3.2. Geographical Named Entity Elements Segmentation

The first subtask is the segmentation of geographic named entity elements. The goal of this task is to ensure the model's predicted segmentation results are as close as possible to the actual annotated results, essentially making it a multi-classification task for each character. In this task, the annotated results are considered a true distribution, while the model's predictions represent an approximate distribution. This is achieved by the model gradually approximating the true annotated data during training, specifically through the minimization of the cross-entropy loss function shown in Equation (1). The model adjusts its predictions based on the annotated data, aiming to make each character's classification prediction closer to the actual label. Specifically, we train the model using a dataset of geographic named entities with labeled categories. With each training iteration, the model adjusts its parameters based on the feedback from the loss function, thereby reducing the discrepancy between its predictions and the actual annotations. As training progresses, the precision of the model's predictions improves, and the final segmentation results increasingly approximate the actual annotated results.

$$Loss_{segmentation} = - \frac{\sum_{i=1}^{batch_size} \sum_{j=1}^C y_j \log P_j}{batch_size} \quad (1)$$

In Equation (1), y_j represents the true category, P_j is the probability distribution output by the model, and C is the number of labeled categories, which in this case is 3. These categories are labeled as follows: "0" and "1" alternately represent geographic named entity elements, while "2" serves as a delimiter represented by the separator "[sep]". The *batch_size* denotes the size of the batch. In addition to evaluating the accuracy of category prediction through Equation (1), we have also introduced metrics such as character-level accuracy, recall, and F1 score to ensure the model performs well at the character level. This multi-scale evaluation can effectively handle cases where the model may miss certain characters or incorrectly segment them. By continuously optimizing model performance, we ensure high precision in identifying and delineating geographic named entity elements in practical applications.

The structure of the geographic named entity elements segmentation module is illustrated in Figure 3. Initially, the decoder input is fed into a fully connected layer for linear transformation to adapt the output dimensions to the target task's labeled categories. Subsequently, the output of this layer is passed into the argmax function, which returns the index of the maximum value along a specified axis. The index of the maximum value output by this function represents the final output category of the annotation. This process is considered a text sequence labeling task, where each character in the text sequence of

geographic named entities is labeled, thereby predicting the boundaries of geographic named entity elements.

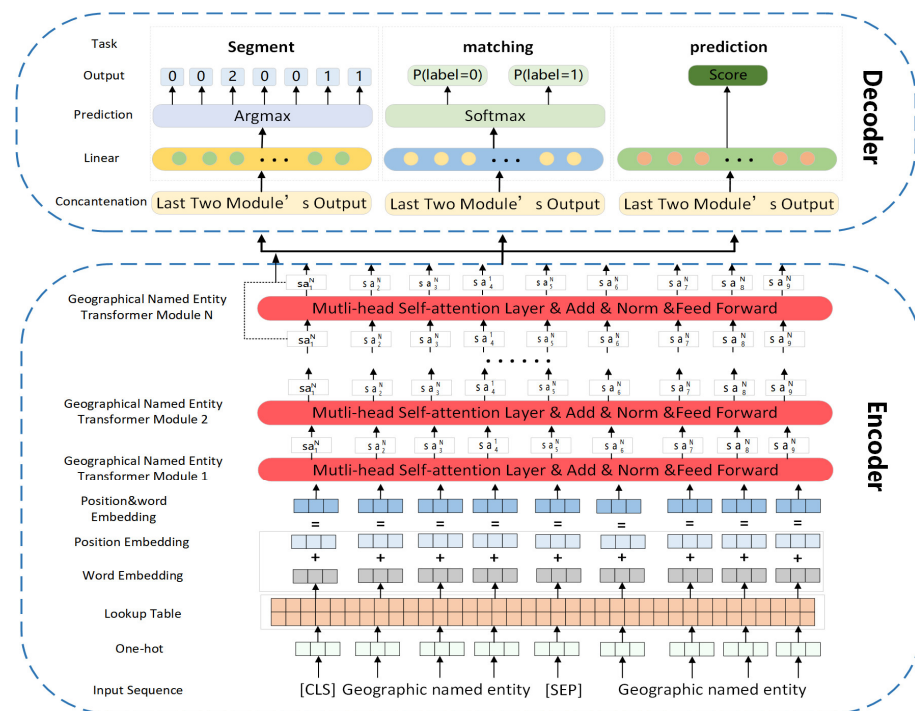


Figure 3. Geographic Named Entity Matching Model.

In the segmentation task, the model undergoes a multi-scale evaluation of character-level accuracy, recall, and F1-score. This rigorous testing not only ensures that the model accurately classifies each character by its category but also correctly identifies the location and label of each character within the phrase. By incorporating these metrics, we address potential issues where the model might incorrectly segment entire phrases or miss some characters. This multi-dimensional assessment allows us to refine the model's performance iteratively, ensuring a high degree of precision in recognizing and delineating geographic named entity elements accurately.

3.3. Geographical Named Entity Spatial–Semantic Matching

The second task involves the geographic named entity spatial–semantic matching task. This task is defined in this study as a binary classification problem, aimed at considering both the spatial location and semantic content of geographic entities. The core task is to determine whether two geographic named entities refer to the same geographic location. This involves assessing not only spatial proximity but also semantic association. Through this approach, we can effectively identify and link entities that match on both geographic and semantic levels, enhancing the model's ability to handle complex geographic data. Unlike traditional address matching, this study introduces a focus on geographic named entity matching without the reliance on a standard geographic named entity database. Consequently, we define this task within the context of this research as “geographic named entity spatial–semantic matching”, emphasizing its unique approach and framework developed specifically for this study. The definition is as follows: for a given geographic named entity dataset $D_{geo_ent} = \{geo_ent_1, geo_ent_2, \dots, geo_ent_n\}$, the goal of the spatial–semantic matching task is to identify pairs of geographic named entities (geo_ent_i, geo_ent_j) that meet specific criteria $geo_ent_i = geo_ent_j$ where $geo_ent_i \in D_{geo_ent}$, $geo_ent_j \in D_{geo_ent}$ and $i \neq j$, where $=$ means both entities refer to the same geographic spatial location.

The method determines whether geo_ent_i and geo_ent_j match by solely utilizing the textual semantic information of the geographic named entities and their corresponding spatial location data, effectively addressing the issue of matching geographic named entities that share the same geographic locations. Additionally, the geographical named entity spatial–semantic matching task defined in this study possesses a symmetric property, meaning if geo_ent_i matches with geo_ent_j , then geo_ent_j also matches with geo_ent_i . The geographical named entity matching module is similar to the feature segmentation module, with the neural network structure shown in Figure 3. Initially, the output of the encoder is passed to a fully connected layer. Since geographical named entity matching is a binary classification task, the decoder’s input is transformed into two dimensions. This process is viewed as a binary classification task, wherein a binary classifier is employed to predict whether a pair of input geographic named entities matches. Subsequently, this layer’s output is fed into a softmax normalization function to calculate the probabilities of whether two geographic named entities match, with specific probabilities $P(\text{label} = 1)$ for a match and $P(\text{label} = 0)$ for a non-match. The objective of this module is to ensure the probability values output by the model closely approximate 1, guided by the true labels, with the loss function defined as follows.

$$Loss_{match} = - \frac{\sum_{i=1}^{batch_size} \log(true_prob[i])}{batch_size} \quad (2)$$

$$true_prob[i] = \text{gather}(prob_i, label_i) \quad (3)$$

In Formulas (2) and (3), $true_prob$ denotes the probabilities of the true labels, $prob$ represents the model’s predicted probability outputs, and $label$ for true label sequence denotes the sequence of true labels. The *gather* method involves aggregating data along a specified dimension according to given indices, i.e., obtaining the predicted values corresponding to the actual labels.

3.4. Geographical Named Entity Spatial Similarity Prediction

The third sub-task is the spatial similarity scoring task. Compared to spatial–semantic matching, spatial similarity scoring focuses on assessing the similarity between entities from the perspective of geographic coordinates. This part primarily addresses geographic entities that are very close in terms of geographic coordinates but have significant differences in semantic text, a situation we refer to as “spatial proximity”. By performing precise similarity calculations on geographic coordinates, we can quantify the degree of spatial closeness between geographic entities, providing more accurate support for subsequent geographic data analysis and applications. Although the initial calculations of spatial similarity prediction are based on traditional geographic distance measurements, the neural network architecture introduced in this study allows us to go beyond simple distance calculations. Neural networks, by integrating spatial information of geographic locations with semantic features of named entities, are able to deeply learn and uncover complex relationships and patterns among geographic entities. This not only enhances the accuracy of similarity scoring but also increases the model’s sensitivity to subtle differences among geographic entities, thereby achieving more precise geographic entity matching in practical applications. Before calculating the spatial similarity of geographic named entities, selecting an appropriate metric is necessary. Given that the dataset in this study is based on geographic coordinates, we have adopted a semi-formalized formula to measure the spatial similarity of geographic named entities, as detailed in Formula (4).

$$d = 2r \arcsin \left(\sqrt{\sin^2 \left(\frac{lat_2 - lat_1}{2} \right) + \cos(lat_1) \cos(lat_2) \sin^2 \left(\frac{lon_2 - lon_1}{2} \right)} \right) \quad (4)$$

In Formula (4), r represents the Earth's average radius, which is set to 6371 km in this study. The term *arcsine* refers to the arcsine function, while *sin* and *cos* represent the sine and cosine functions, respectively. lon_1 and lon_2 denote the longitude of geographical named entity data points, while lat_1 and lat_2 represent the latitude of points in the geographic named entity dataset. Formula (4) is employed to compute the spatial distance between pairs of geographic locations. However, the role of the neural network depicted in Figure 3 extends beyond simple distance calculations. It integrates the spatial distance with other features, such as semantic information, to generate a more comprehensive geographic named entity similarity score. This multitask learning framework does not rely solely on geographic distances but also leverages the neural network model to better capture and integrate spatial and semantic information, thus enhancing the overall matching performance.

For the final score prediction, we measure the spatial similarity between geographical named entities using a scoring method. The computation process is as follows, with the $score_{max}$ value set to 5 in this study.

(1) Define the maximum score for spatial similarity, referred to as $score_{max}$, which is the spatial similarity score when two geographical named entities have exactly the same geographical location.

(2) Calculate the maximum distance d_{max} between geographic named entities in the dataset; when the distance d_{max} between two entities is reached, their spatial similarity score is 0.

(3) Calculate the distance between geographic named $entity_i$ and geographic named $entity_j$, denoted as $d_{\{i,j\}}$.

(4) Compute the spatial similarity score between geographic named entities.

$$score_{i,j} = score_{max} - \frac{score_{max} * d_{i,j}}{d_{max}} \quad (5)$$

(5) Repeat steps (3) and (4) until the spatial similarity calculations for all geographic named entities are completed.

3.5. Geographical Named Entity Matching Model

The encoding component of GNEMM employs an optimized BERT framework to deeply analyze the semantic information of geographic named entity texts, utilizing a multi-head self-attention component at the end after removing the original output part. This component enhances the extraction of semantic features significantly through multiple stacked self-attention modules, designed to refine the understanding of complex semantic relationships within geographic texts. Additionally, the model incorporates residual connections, layer normalization and a fully connected feed-forward network, adhering to the Transformer architecture. As the model processes pairs of geographic named entities, the input text sequence is modified by inserting a [CLS] token at the beginning of the sequence to serve as a sentence-initial marker, which then serves as the input for the subsequent "decoder". The traditional Transformer decoder component was omitted to better suit the specifics of our geographic named entity corpus and the associated pre-training tasks. The decoder in GNEMM includes three subtasks, and this study balances the loss of each through a weighting of the three subtasks' importance. A weighted average method is employed to calculate the model's final loss function, with the formula presented as Equation (5).

$$Loss_{total} = w_1 * Loss_{segmentation} + w_2 * Loss_{match} + w_3 * Loss_{score} \quad (6)$$

In Formula (6), w_1, w_2 and w_3 represent the weight coefficients for each task, with specific values determined through an analysis of experimental results. The neural network framework of GNEMM is illustrated in Figure 3.

3.6. Geographical Named Entity Recommendation Based on Retrieval-Augmented Generation

The RAG-based address recommendation process consists of three stages, as illustrated in Figure 4. The first stage is retrieval, aimed at extracting the most relevant address data from a large dataset. Initially, an external knowledge base is selected and vectorized, utilizing 63,000 new addresses and related information from 2022 to 2023, gathered to create a comprehensive and structured repository of geographic information. This database was specifically constructed to support the nuanced demands of spatial–semantic matching tasks in this study. This knowledge base includes fields such as address, name, administrative area, and coordinates, specifically designed to provide a structured source of geographic information to support deep matching tasks. Each address and its related information are vectorized as a document block, using the faiss indexing library to link metadata with the vector library, employing OpenAI’s embedding method to generate the external vector library. This setup not only allows for efficient retrieval of relevant information but also enables the integration of real-time geographic data, enhancing the model’s ability to address non-standardized data challenges. Upon user query, the input is vectorized in the same manner and searched in the vector library to return the most matching address and related information. In this study, the retrieval stage is set to return three address document blocks, which are the closest to the query address. We validated the effectiveness of different vectorization methods in retrieving and matching various types of geographical named entities, using the distance and standard deviation between the query and matched addresses as metrics. The calculations are consistent with Equation (4).

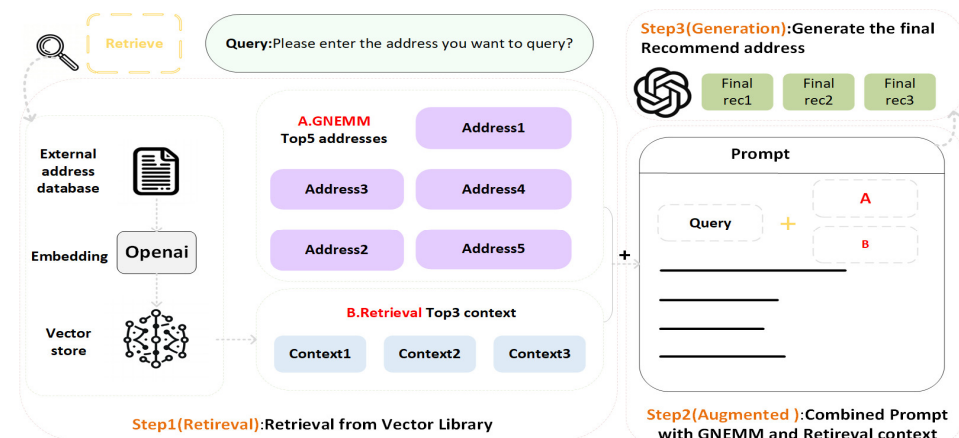


Figure 4. Combining the Retrieval Augmented Generation result process with GNEMM.

The Augmented stage aims to integrate the retrieved information. Unlike traditional RAG, which simply combines retrieved information with prompts, we additionally incorporate matched geographical named entities from the GNEMM with the extra geographical named entity information obtained during the retrieval stage, using a Chain of Thought prompt strategy. This strategic integration uses insights derived from logical sequences similar to expert analysis, thereby enhancing the model’s decision-making process. The Chain of Thought strategy employed optimizes the model’s responses by simulating the human problem-solving process. Specifically, the RAG module not only answers direct queries but also generates explanations for its answers in intermediate steps, akin to the logical sequences considered by humans when solving problems. This approach, compared

to direct retrieval or simple prompts, provides deeper semantic understanding, enabling the model to generate more accurate and reasonable predictions. Additionally, while other prompting strategies were considered, the Chain of Thought method was found to be particularly effective in utilizing the extensive information from the external knowledge base for complex tasks involving geographical named entities. This step primarily combines the top five matching geographical named entities from the GNEMM with three related pieces of geographical named entity information from the retrieval stage, constructing a complete prompt content in a Chain of Thought manner, serving as input for the final generation stage.

In the generation stage, we utilize current large language models such as GPT-4-Turbo, GLM-3, and Claude3 to perform generation tasks. The input is the integrated prompt content obtained in the Augmented stage. Given the powerful semantic understanding capabilities of the GPT-4-Turbo model, we select it as the generative model for the final geographical named entity recommendation task. This approach allows the model to leverage both spatial and semantic information, reflecting the integration strategy designed to maximize the accuracy and relevance of the recommendations. It is worth mentioning that the purpose of using LLM in this study is to explore its ability to understand geography conveniently in the presence of an external knowledge base, which is different from traditional geocoding. This exploration opens up new directions for future research and applications.

4. Experiments Results and Discussion

This section will provide a comprehensive report on the experimental results of the proposed Geographical Named Entity Matching Model and the application of Retrieval-Augmented Generation (RAG) technology, along with a detailed discussion of the relevant findings.

4.1. Multi-Objective Geographical Named Entity Matching

To determine the weighting coefficients for each task within the loss function of the Geographical Named Entity Matching Model, this study initially set multiple weight combinations for sensitivity analysis to select the optimal performance weight combination. We established six different sets of weight combinations in experiments, with the parameter settings for each sub-task presented in Table 5. The loss curves for the various weight combinations are illustrated in Figure 5. The selection of specific weight coefficients was guided by a sensitivity analysis, recognizing that each subtask contributes differently to the overall performance of the model. The weight settings are intended to balance these contributions to optimize the model's overall effectiveness. Therefore, six different weight configurations were tested, each reflecting varying levels of emphasis on the tasks' impact on the loss function. This approach allows for targeted adjustments based on the specific influence of each task within the overall model structure.

Table 5. Different subtask weight coefficient combinations.

Combination Round	w_1	w_2	w_3
1	0.1	0.1	0.8
2	0.2	0.2	0.6
3	0.3	0.3	0.4
4	0.4	0.4	0.2
5	0.2	0.6	0.2
6	0.1	0.8	0.1

Figure 5 shows a rapid decline in loss values before 200 k, followed by a slow downward trend and a final stabilization. An early stopping strategy was implemented during training, and hence the iteration steps and timing under different weight coefficients varied, but ultimately the loss values decreased to a low level, demonstrating that the model was adequately trained. Additionally, to select the optimal weight combination, multiple evaluation metrics were used. The evaluation metrics for different weight combinations are presented in Table 6.

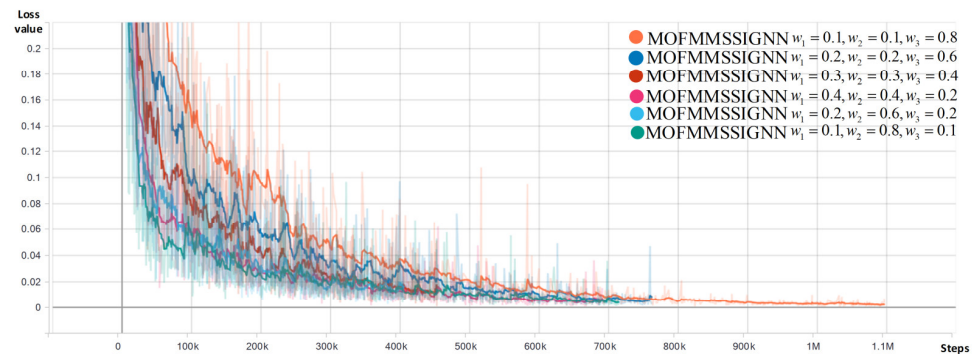


Figure 5. Geographic Named Entity Matching Model training in different combinations of multi-objective weights.

Table 6. Evaluation metrics for different weight coefficient combinations.

Weight Coefficient	Element Segmentation Precision	Element Segmentation Recall	Element Segmentation F1 Score	Match Precision	Match Recall	Match F1 Score	Spatial Similarity Prediction R ²
1	0.9026	0.9809	0.9392	0.9989	0.9896	0.9942	0.8076
2	0.8018	0.9796	0.8809	0.9980	0.9912	0.9946	0.7748
3	0.8579	0.9860	0.9167	0.9981	0.9860	0.9920	0.8150
4	0.8701	0.9819	0.9215	0.9984	0.9895	0.9939	0.7459
5	0.8974	0.9837	0.9375	0.9986	0.9906	0.9946	0.8028
6	0.8986	0.9794	0.9366	0.9981	0.9916	0.9948	0.4745

Bold indicates the best result.

Table 6 indicates no significant correlation between the magnitude of weight coefficients and the corresponding task's evaluation metrics. For instance, increasing the weight of the loss function for geographical named entity matching did not significantly improve the evaluation metrics for that task. This also demonstrates the interplay among subtasks during multi-task learning. When the weight combination for the loss function was set to ($w_1 = 0.1$, $w_2 = 0.1$, $w_3 = 0.8$), the model maintained high accuracy in the geographical named entity matching task and performed well in the geographical named entity element segmentation and spatial similarity score prediction tasks, making this weight combination the basis for subsequent experiments.

To assess the effectiveness of subtask modules within the model, we conducted ablation experiments for each subtask, with corresponding accuracy metrics shown in Table 7. Initially, the model, containing only the geographical named entity matching module, achieved high levels on relevant evaluation metrics with an F1 score of 0.98, demonstrating a strong semantic understanding of geographical named entities. Adding geographical named entity element segmentation or spatial similarity score prediction modules to the basic matching task improved performance in geographical named entity matching, indicating that these submodules positively influenced the overall task. Integrating all three task modules resulted in the best overall performance, with increases in matching precision and F1 scores and significant improvements in element segmentation accuracy. However, the recall rates for element segmentation and matching slightly decreased. Notably, the

performance of models with combinations of geographical named entity element segmentation + matching and matching + spatial similarity score prediction was relatively poorer compared to the integrated three-task module model, confirming the interdependent and facilitative relationships among the subtasks.

After verifying the effectiveness of each sub-module, we further conducted a comprehensive comparison of our matching model with widely recognized traditional models in the field of geographic named entity matching, such as BERT, ELECTRA, and ERNIE. The related results are displayed in Table 8. Overall, all models performed well on the matching metrics, with our proposed GNEMM model showing notable improvements in several key areas. In the task of geographical named entity element segmentation, our model achieved a significant advantage, with a segmentation precision of 0.9026 and an F1 score of 0.9392, outperforming the other models by a considerable margin. While models like ELECTRA and ERNIE also demonstrated strong segmentation recall (0.9745 and 0.9795, respectively), their precision and F1 scores were notably lower than GNEMM. This suggests that our model is more effective in correctly identifying the exact boundaries of geographical named entities, particularly in more complex cases where other models may struggle. In terms of matching performance, all models achieved high precision and recall, with ELECTRA showing slightly higher recall (0.9900) compared to our model's 0.9896. However, our GNEMM model still outperformed the others in terms of overall match score (0.9942) and F1 score (0.9941), indicating a more balanced and robust performance across different matching tasks. One of the most significant improvements of our model was in the spatial similarity prediction task, where GNEMM achieved an R^2 score of 0.8076, substantially higher than the other models. This demonstrates our model's superior ability to capture the spatial hierarchical relationships and the relative positions of geographic named entities, which are crucial for tasks that require a deep understanding of spatial contexts. In conclusion, while traditional models like BERT, ELECTRA and ERNIE have proven effective for various natural language tasks, our proposed GNEMM model exhibits clear advantages in geographical named entity element segmentation and spatial similarity prediction, making it a more suitable choice for complex geographic named entity matching tasks.

Table 7. Fusion experiments of different subtask combinations.

Task Combination	Geographic Named Entity Matching	Geographic Named Entity Element Segmentation + Geographic Named Entity Matching	Geographic Named Entity Matching + Geographic Named Entity Spatial Similarity Prediction	Geographic Named Entity Element Segmentation + Geographic Named Entity Matching + Geographic Named Entity Spatial Similarity Prediction
Element Segmentation Precision	/	0.6261	/	0.9026
Element Segmentation Recall	/	0.9818	/	0.9809
Element Segmentation F1 Score	/	0.7416	/	0.9392
Match Precision	0.9799	0.9825	0.9817	0.9989
Match Recall	0.9798	0.9824	0.9904	0.9896
Match F1 Score	0.9798	0.9824	0.986	0.9942
Spatial Similarity Prediction R^2	/	/	0.7638	0.8076

Bold indicates the best result.

Table 8. Comparison experiments of different matching models.

Model Name	Element Segmentation Precision	Element Segmentation Recall	Element Segmentation F1 Score	Match Precision	Match Recall	Match F1 Score	Spatial Similarity Prediction R^2
BERT	0.2568	0.9174	0.3940	0.9834	0.9719	0.9776	0.3622
ELECTRA	0.7970	0.9745	0.8749	0.9969	0.9900	0.9934	0.6726
ERNIE	0.7203	0.9795	0.8289	0.9979	0.9894	0.9936	0.7547
GNEMM	0.9026	0.9809	0.9392	0.9989	0.9896	0.9942	0.8076

Bold indicates the best result.

4.2. Geographical Named Entity Recommendation Based on Retrieval-Augmented Generation

In the RAG-based geographical named entity recommendation tasks, we constructed an external knowledge base using newly added Shandong address information from 2023, sourced from a government-collected standard address dataset that includes geographic named entity data added over the last two years. Unlike the data used for training the GNEMM, this external knowledge base is specifically employed for RAG experiments to support and augment the model’s functionality. We carefully selected key fields including the names, addresses, aliases, and geographical coordinates of entities, which are critical for conducting high-precision geographic information retrieval. This selection ensures that the RAG module can effectively extract relevant supplementary information from the external knowledge base. Through this approach, the RAG module can more comprehensively understand and handle the relationships and geographical connections between entities, significantly enhancing the model’s applicability and accuracy in complex geographic environments. We then generated the required vector library using OpenAI’s embedding technology. In the Augmented phase, we employed a Chain of Thought prompt strategy, with detailed prompts shown in Table 9 and specific examples illustrated in Figure 6. The generative model for geographical named entity recommendations was performed using gpt-4-turbo, with related experimental parameters detailed in Table 10.

Table 9. Retrieval-Augmented Generation’s prompts.

Prompt Strategies	Prompt Content
Chain of Thought	Task Description: Recommend three geographical named entities most relevant to a given query from a list of candidate geographical named entities, based on their proximity and relevance to the query. User’s query geographical named entity: (query), with latitude and longitude of the query geographical named entity: query_lat_lon1. Top 5 matching geographical named entity and their latitudes and longitudes provided by our model: [GNEMM_lat_lon1] Relevant information retrieved from the external address database using the RAG method: (content). Please analyze step by step according to the following steps: Step 1: Extract geographical named entity content related to the query address from the RAG output. Step 2: Analyze the user query to identify key geographical or notable terms of the query geographical named entity. Step 3: Filter out geographical named entities that contain the identified keywords from both the top 5 geographical named entities provided by the model and geographical named entities extracted from the RAG content. Step 4: Calculate the geographical distance between each geographical named entity and the query geographical named entity using their latitude and longitude coordinates. Step 5: Sort the filtered geographical named entities based on their proximity to the query geographical named entity and how well they match the identified keywords. Step 6: Select the top three geographical named entities as the recommendation based on their ranking from the previous step. Step 7: Ensure the query geographical named entity is included among the top three recommended geographical named entities if it appears in the list of candidate addresses. Step 8: Output the three recommended geographical named entities along with their geographical coordinates as the final output.

Table 10. RAG experimental parameters.

Experimental Parameter	Index
Address Dataset entry 2	2023 newly added Shandong address data data
Vectorization Method	OpenAI embedding
Large Language Model	gpt-4-turbo
Prompt Strategy	Chain of Thought (COT)
Evaluation Parameter 1	Average Distance Difference
Evaluation Parameter 2	Standard Deviation of Average Distance

In this study, we conducted research on geographical named entity recommendation tasks using the GNEMM model. To assess model performance, we calculated the average distance difference, median, and minimum values between multiple matched addresses

and the query address. Table 11 displays the comparison results of the GNEMM model with other benchmark models (BERT-Based, ELECTRA-Based and ERNIE-Based). Here, the K-value represents the number of geographical named entities matched by the model. Table 11 lists the average distance differences, median, and minimum values for different models at K-values of 1, 3, and 5, providing a comprehensive assessment of each model's performance in geographical named entity matching tasks.



Figure 6. Example of processing RAG using the Chain of Thought prompt strategy in gpt4.

Table 11. Distance error mean value for different models.

Model Name	Distance Error Mean Value Index K Value	Metric		
		Average	Median	Minimum
BERT_Based	1	11.4794	11.3948	0.6935
	3	11.0118	10.9464	5.7863
	5	10.6615	10.2863	6.3710
ELECTRA_Based	1	10.2048	11.2804	0.7098
	3	9.9437	10.2734	3.8479
	5	9.7491	10.1503	6.1102
ERNIE_Based	1	8.9569	9.9712	0.4891
	3	8.8730	7.3899	1.2030
	5	8.7860	6.6467	3.8419
GNEMM (ours)	1	2.0057	1.9718	0
	3	2.0236	1.9086	0.0181
	5	2.0365	1.8896	0.0160

Bold indicates the best result.

From Table 11, it is evident that as the K-value increases, the average and median values for the control models show a decreasing trend, while our matching model maintains an average value of around 2 km, indicating little overall change. This suggests that as the number of recommended addresses increases, the distance between the query address and recommended addresses decreases, and due to the high precision of our matching model, the distance deviation is minimal. The model proposed in this study achieves a minimum average distance error of 2.0236 km, significantly lower than the minimum average value of the control models, which is 3.4 times greater. Regarding the median error, the median of our model is 1.89 km, markedly lower than the 6.65 km of the control models. Finally, the minimum of the average distance errors can be used to assess the

best recommendation results of the model. Overall, all models achieved relatively low minimums in the recommendation tasks, sequentially reaching 0.69 km, 0.70 km, 0.48 km, 0.17 km and 0 km. Compared to other control models, the minimum value of our model is the lowest, demonstrating its excellent performance in recommendation tasks. Notably, the lowest value reached 0 km, a rare occurrence, indicating that the queried geographical named entity exists within the standard address database.

Subsequently, we conducted geographic named entity recommendation tasks based on RAG technology. During the retrieval phase of RAG, we evaluated the effectiveness of address vectors generated using TF-IDF and those generated by the GNEMM in retrieving various types of geographical named entities. The results are illustrated in Figure 7.

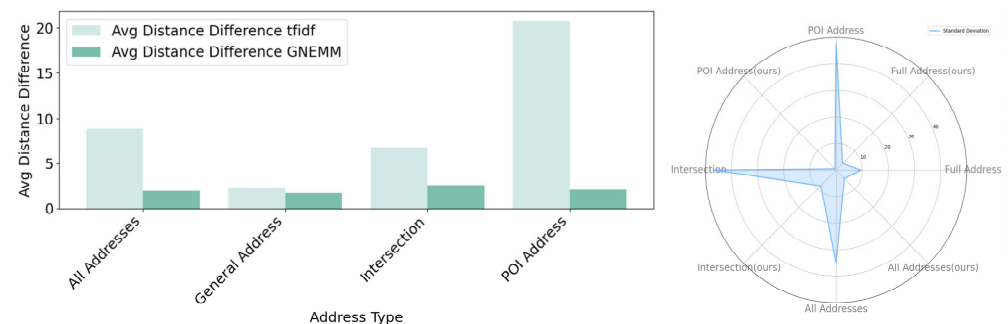


Figure 7. The average distance difference and standard deviation of the top five matching geographical named entities for different types of geographical named entities.

The bar graph on the left side of Figure 7 depicts the average distance differences for various types of query addresses in retrieving and matching geographical named entities. This metric reflects the distance between the matched and query geographic named entities, where the smaller the distance, the closer the retrieved and matched geographic named entity are to the query address. The radar chart on the right shows the average distance standard deviations for various types of addresses during the matching process. This value measures the stability of match results, where the smaller the standard deviation, the higher the stability of the matched geographical named entity. From the average distance metrics, our multi-objective model provides geographical named entity matching results across all types of addresses with smaller distance differences compared to traditional vector methods, especially for point-of-interest addresses, where the difference is about one-tenth of that of traditional methods, demonstrating high efficiency in resolving specific geographical named entities. Regarding the standard deviation, our model exhibits significantly lower standard deviations in matching geographical named entities compared to traditional methods, particularly at intersection addresses, with stability about 20 times higher, further demonstrating the model's high stability in geographical named entity matching tasks.

We conducted a geographic named entity recommendation task using the RAG method, demonstrating six instances across three types of geographic named entities. The first type comprises complete geographic named entities, which include detailed information such as building numbers, street names, districts, cities, states/provinces, countries, and postal codes. The second type, points of interest (POIs), refers to locations of specific significance and function, such as hospitals and schools. The third type is geographic named entities for intersections, indicating the geographic positions where two or more roads intersect. To clearly demonstrate the recommendation results for these geographic named entities, we used the “query” label to uniformly identify different types of query entities. The top five geographic named entities matched by GNEMM are labeled GNEMM1 to GNEMM5, while those retrieved by the RAG model are denoted as RAG1 to

RAG3. Figure 8 displays these visualization results, with the RAG results in Figure 8a–f corresponding specifically to the content in Tables 12–17.



Figure 8. RAG recommendation results for different types of geographical named entities.

In the case of the complete geographic named entity “42 Yihe Village, Gucheng Street, Shouguang City, Weifang City”, the RAG model’s recommendations are shown in Figure 8a and detailed in Table 12, listing the top five geographic named entities matched by GNEMM and entities retrieved externally by RAG. Upon detailed analysis, the GPT-4 model ultimately recommended “26, 29 and 27 Yihé Village, Gucheng Subdistrict, Shouguang City, Weifang City”, all of which were matched by GNEMM. During the analysis, the GPT-4 model identified key geographic named entity elements within the query address, including “Weifang City”, “Shouguang City”, “Gucheng Subdistrict” and “Yihé Village”, establishing a four-tier geographic named entity framework. The top five geographic named entities matched by GNEMM align perfectly with the query address in terms of hierarchy, differing only in specific house numbers. The geographic named entities retrieved by RAG include provincial elements, which do not match the hierarchy as precisely as the GNEMM results. Spatially, the geographic named entities matched by GNEMM and retrieved by RAG are located within the same village and are very close to each other. Consequently, the GPT-4 model selected the geographic named entities with more precise hierarchical matches “26, 29 and 27”, which are not only spatially close to the queried address but also fully consistent with the hierarchy of geographic named entities, aligning with the user’s address usage logic and habits. This demonstrates the precision and effectiveness of our model in considering spatial proximity and hierarchical matching, emphasizing the importance of precise hierarchical alignment in address recommendations. It also indicates that although RAG technology adds additional address information to the recommendation of geographic named entities, it does not significantly improve the final recommendation results.

Table 12. Complete geographical named entity1 RAG results.

Query	GNEMM's Matching Geographical Named Entity	Distance	RAG's Geographical Named Entity	Distance	Recommend Geographical Named Entity
42 Yihe Village, Gucheng Street, Shouguang City, Weifang City	GNEMM1: 26 Yihe Village, Gucheng Street, Shouguang City, Weifang City	0.065	RAG1: 36 Yihe Village, Gucheng Street, Shouguang City, Weifang City, Shandong Province	0.169	26 Yihe Village, Gucheng Street, Shouguang City, Weifang City
	GNEMM2: 29 Yihe Village, Gucheng Street, Shouguang City, Weifang City	0.026	RAG2: 98 Yihe Village, Gucheng Street, Shouguang City, Weifang City, Shandong Province	0.169	29 Yihe Village, Gucheng Street, Shouguang City, Weifang City
	GNEMM3: 38 Yihe Village, Gucheng Street, Shouguang City, Weifang City	0.083	RAG3: 153 Yihe Village, Gucheng Street, Shouguang City, Weifang City, Shandong Province	0.169	27 Yihe Village, Gucheng Street, Shouguang City, Weifang City
	GNEMM4: 27 Yihe Village, Gucheng Street, Shouguang City, Weifang City	0.048			
	GNEMM5: 14 Yihe Village, Gucheng Street, Shouguang City, Weifang City	0.103			

Table 13. POI geographical named entity1 RAG results.

Query	GNEMM's Matching Geographical Named Entity	Distance	RAG's Geographical Named Entity	Distance	Recommend Geographical Named Entity
Shandong Art Academy, Jinan	GNEMM1: Longhu Jinan Olympic Sports Street	0.92	RAG1: 5th Building, 91 Cultural Road, Lixia District, Jinan City, Shandong Province	5.326	5th Building, 91 Cultural Road, Lixia District, Jinan City, Shandong Province
	GNEMM2: 61 Industrial South Road	2.679	RAG2: 91 Cultural East Road (Shandong Art Academy, East Campus)	5.421	91 Cultural East Road (Shandong Art Academy, East Campus)
	GNEMM3: 60 Industrial South Road	2.351	RAG3: 91 Cultural East Road	5.429	RAG3: 91 Cultural East Road
	GNEMM4: 83 Industrial South Road	2.111			
	GNEMM5: 90 Industrial South Road	2.441			

Red font indicates addresses with excessive error distance.

Table 14. Intersection geographical named entity1 RAG results.

Query	GNEMM's Matching Geographical Named Entity	Distance	RAG's Geographical Named Entity	Distance	Recommend Geographical Named Entity
100 m to the north, west, and south of the intersection of Jingzhong Road and Jinghua Street	GNEMM1: 120 m to the north, west, and south of the intersection of Jingzhong Road and Jinghua Street	0.026	RAG1: 160 m due south of the intersection of Jingzhong Road and Jinghua Street	0.117	120 m to the north, west, and south of the intersection of Jingzhong Road and Jinghua Street
	GNEMM2: 30 m due south of the intersection of Jingzhong Road and Jinghua Street	0.082	RAG2: 80 m due south of the intersection of Jingzhong Road and Jinghua Street	0.15	30 m due south of the intersection of Jingzhong Road and Jinghua Street
	GNEMM3: 70 m due south of the intersection of Jingzhong Road and Jinghua Street	0.095	GNEMM3: 70 m due south of the intersection of Jingzhong Road and Jinghua Street	0.164	70 m due south of the intersection of Jingzhong Road and Jinghua Street
	GNEMM4: 50 m to the southeast of the intersection of Jingzhong Road and Jinghua Street	0.100			
	GNEMM5: Due north of the intersection of Jingzhong Road and Jinghua Street	0.106			

Table 15. Complete geographical named entity2 RAG results.

Query	GNEMM's Matching Geographical Named Entity	Distance	RAG's Geographical Named Entity	Distance	Recommend Geographical Named Entity
4th Floor, Zhongrui International, No. 119 Qinhuangdao Road, Donggang District, Rizhao City, Shandong Province	GNEMM1: No. 567 Dalian Road, Donggang District, Rizhao City, Shandong Province	1.293	RAG1: 4th Floor, Zhongrui International, No. 119 Qinhuangdao Road	0	4th Floor, Zhongrui International, No. 119 Qinhuangdao Road
	GNEMM2: No. 1 Jiaozhou Road, Donggang District, Rizhao City, Shandong Province	2.581	RAG2: No. 119 Qinhuangdao Road, Donggang District, Rizhao City and Jinghua Street	0	No. 119 Qinhuangdao Road, Donggang District, Rizhao City
	GNEMM3: No. 14 Tianjin Road, Donggang District, Rizhao City, Shandong Province	1.697	RAG3: Zhongrui International, Qinhuangdao Road, Donggang District, Rizhao City	0.001	Zhongrui International, Qinhuangdao Road, Donggang District, Rizhao City
	GNEMM4: Donggang District, Rizhao City, Shandong Province	1.573			
	GNEMM5: No. 68 Gucheng Road, Donggang District, Rizhao City, Shandong Province	12.266			

Table 16. POI geographical named entity2 RAG results.

Query	GNEMM's Matching Geographical Named Entity	Distance	RAG's Geographical Named Entity	Distance	Recommend Geographical Named entity
Qushuiting Street, Jinan, Qushuiting No.	GNEMM1: Qushuiting Street Community	0.64	RAG1: No. 27 Qushuiting Street, Jinan	0.68	Qushuiting Street Community
	GNEMM2: Next to Fuxue Confucian Temple, Daming Lake Road, Jinan	0.75	RAG2: No. 5 Qushuiting Street, Lixia District, Jinan City, Shandong Province	0.85	GNEMM3: No. 10 Qushuiting Street
	GNEMM3: No. 10 Qushuiting Street	0.68	RAG3: No. 116 Houzaimen Street and Jinghua Street	0.73	No. 27 Qushuiting Street, Jinan
	GNEMM4: Junyi Zuoan, No. 18 Xiluohe Road	2.83			
	GNEMM5: Zhijinshi Street Community	1.05			

Table 17. Intersection geographical named entity2 RAG results.

Query	GNEMM's Matching Geographical Named Entity	Distance	RAG's Geographical Named Entity	Distance	Recommend Geographical Named Entity
462 m east of the intersection of Yuehe Road and Xuanwu West Street, on the north side of the road	GNEMM1: 451 m east of the intersection of Yuehe Road and Xuanwu West Street, on the north side of the road	0.011	RAG1: 462 m east of the intersection of Yuehe Road and Xuanwu West Street, on the north side of the road	0	451 m east of the intersection of Yuehe Road and Xuanwu West Street, on the north side of the road
	GNEMM2: 132 m east of the intersection of Yuehe Road and Xuanwu West Street, on the north side of the road	0.33	RAG2: 150 m east of the intersection of Xuanwu West Street and Yiyuan Road, on the south side of the road	1.59	539 m east of the intersection of Yuehe Road and Xuanwu West Street, on the north side of the road
	GNEMM3: 539 m east of the intersection of Yuehe Road and Xuanwu West Street, on the north side of the road	0.077	RAG3: 180 m due east of the intersection of Nanguan Street and Nanhu Bin Road, and 150 m to the east	348.56	462 m east of the intersection of Yuehe Road and Xuanwu West Street, on the north side of the road
	GNEMM4: 110 m east of the intersection of Yuehe Road and Xuanwu West Street, on the north side of the road	0.352			
	GNEMM5: 492 m east of the intersection of Youai Road and Xuanwu West Street, on the north side of the road	1.188			

Red font indicates addresses with excessive error distance.

For the point of interest “Shandong Art Academy, Jinan” the RAG model’s recommendations are spatially displayed in Figure 8b. The specific matching and retrieval results are detailed in Table 13. After thorough analysis, the GPT-4 model ultimately recommended three geographic named entities, “5th Building, 91 Cultural Road, Lixia District, Jinan City, Shandong Province”, “91 Cultural East Road (Shandong Art Academy, East Campus)” and “91 Cultural East Road”. Although the entities matched by GNEMM were spatially closer to the queried address, GPT-4 selected three entities retrieved by RAG that were comparatively farther away. Semantically, GPT-4 identified the key geographic named entity element “Shandong Academy of Arts” within the query address, while the top five entities matched by GNEMM lacked similar semantic elements. Although the geographic named entities retrieved by RAG were farther from the query address, they were semantically strongly linked to it, such as “91 Cultural East Road (Shandong Art Academy, East Campus)”, which contains the key geographic named entity element. Further analysis of the final recommendations revealed discrepancies in spatial information between the current address and the data in the standard address database used to train the GNEMM model, leading to inconsistencies in spatial positioning, even though these matches were semantically inconsistent. This indicates that the GPT-4 model, similar to human cognition, considers not only spatial elements but also semantic information when recommending geographic named entities. Furthermore, the RAG approach has revealed some changes in addresses, helping us to correct spatial or semantic errors in geographic named entities.

For the query “100 m to the north, west, and south of the intersection of Jingzhong Road and Jinghua Street” the recommended results are spatially displayed in Figure 8c, with specific matches and retrieval results listed in Table 14. The GPT-4 model conducted an in-depth analysis of the user query, identifying key geographic elements of the query address, including “Jingzhong Road”, “Jinghua Road”, “intersection”, “northwest direction” and “100 m”. Considering the uniqueness of intersection-type geographic named entities, the GPT-4 model primarily focused on matching key elements in its recommendations. The final recommended geographic named entities were “120 m to the north, west, and south of the intersection of Jingzhong Road and Jinghua Street”, “30 m due south of the intersection of Jingzhong Road and Jinghua Street”, and “70 m due south of the intersection of Jingzhong Road and Jinghua Street”, all of which are consistent or close to the direction of the query address. As intersection-type geographic named entities do not include traditional geographic elements such as provinces, cities, or districts, the GPT-4 model made precise considerations regarding direction and distance. In selecting recommended geographic named entities, priority was given to those that align with the direction of the query address. If no geographic named entities in the same direction exist, the GPT-4 model considers other directions and distances to ensure that the recommended entities are as close as possible to the query address.

The spatial recommendation results for the address “4th Floor, Zhongrui International, No. 119 Qinhuangdao Road, Donggang District, Rizhao City, Shandong Province” are displayed in Figure 8d. Related detailed matching results are listed in Table 15. In the geographic named entities matched by GNEMM, the closest is approximately 1.29 km from the query address, while the farthest exceeds 12 km. In contrast, among the geographic named entities retrieved by RAG from an expanded external address database, two overlap with the query address, and one is very close. GPT-4, after analyzing the key elements and spatial distances of these geographic named entities, recommended three entities, “4th Floor, Zhongrui International, No. 119 Qinhuangdao Road”, “No. 119 Qinhuangdao Road, Donggang District, Rizhao City” and “Zhongrui International, Qinhuangdao Road, Donggang District, Rizhao City”. These three recommended entities, all sourced from additional address information retrieved by RAG, are highly relevant to the query

address, despite differing in presentation. For instance, “4th Floor, Zhongrui International, No. 119 Qinhuangdao Road” lacks the province, city, and district prefix. “No. 119 Qinhuangdao Road, Donggang District, Rizhao City” lacks provincial, building name and floor information. “Zhongrui International, Qinhuangdao Road, Donggang District, Rizhao City” lacks provincial and road-level geographic named entity elements. Meanwhile, the top five geographic named entities matched by GNEMM only succeeded at the district level and failed to match similar geographic named entities at deeper hierarchical levels. Therefore, after a comprehensive analysis of the detailed semantics and spatial distances of the geographic named entities, GPT-4 selected these three entities for the final recommendation. This reflects that the GPT-4 model, in the recommendation process, not only values the hierarchy of geographic named entities but also considers their spatial proximity to the query address.

The RAG results for the POI address “Qushuiting Street, Jinan, Qushuiting No.” are spatially displayed in Figure 8e. Specific matches and retrieval results are listed in Table 16. After in-depth analysis, GPT-4 recommended three geographic named entities, “Qu Shui Ting Street Community”, “No. 10 Qushuiting Street” and “No. 27 Qushuiting Street, Jinan”. During the recommendation process, GPT-4 considered the common geographic named element “Qu Shui Ting” shared by all three recommended entities. Additionally, although “Next to Fuxue Confucian Temple, Daming Lake Road, Jinan” ranked second in GNEMM matching, it was not included in the final recommendations due to its lower relevance of geographic elements and greater spatial distance. Notably, although “No. 5 Qushuiting Street, Lixia District, Jinan City, Shandong Province” shares geographic named elements such as “Jinan” and “Qu Shui Ting” with the query address, it was not recommended due to its relatively greater distance. This indicates that the GPT-4 model comprehensively considers semantic and spatial information in the matching and recommendation of geographic named entities.

Figure 8f shows the spatial distribution of RAG results for the query intersection “462 m east of the intersection of Yuehe Road and Xuanwu West Street, on the north side”. Specific matching and retrieval results can be found in Table 17. After detailed analysis, GPT-4 ultimately recommended three entities, “462 m east of the intersection of Yuehe Road and Xuanwu West Street, north side”, “451 m east of the intersection” and “539 m east of the intersection”. Among these, “462 m east of the intersection of Yuehe Road and Xuanwu West Street, north side” was directly listed as a recommended address due to its perfect match with the query address. GNEMM’s recommendations primarily considered spatial proximity to the query address, particularly the first three geographic named entities, all of which prominently included the key geographic element “east of the intersection of Yuehe Road and Xuanwu West Street”. Therefore, after integrating geographic element matching and spatial proximity, GPT-4 selected the two closest addresses for recommendation. Additionally, an anomaly was observed in the information returned from the RAG retrieval; although the geographic named entity contained the element “east side of the intersection”, it was completely different from the actual queried intersection, with a spatial difference of up to 348 km, likely due to limitations in the external address database.

In the RAG-based geographical named entity recommendation tasks, we analyzed the recommendations for various types of geographical named entities. Comparing the final results from Figure 8a–c, we found that the RAG method does not necessarily enhance recommendation outcomes. Although RAG technology enhances the diversity of geographical named entities by incorporating an external knowledge base, its primary role is to supplement and extend existing Geographical Named Entity Matching Models rather than replace them. Sometimes, newly added external knowledge can confuse the results of the original model, as mentioned in the analysis of Figure 8b, where the initially

queried geographical named entity had spatial discrepancies with the latest geographical named entity information, leading to mismatches in the matching model. The generative model of RAG, through in-depth analysis and consideration, mitigated the consequences of mismatches, correctly recommending suitable geographical named entities to users. Additionally, the model performs well when matching intersection-type geographical named entities, considering not only spatial distance but also directional accuracy, ensuring the practicality and accuracy of the recommendations. The experimental results from Figure 8c demonstrate the challenges of intersection address recommendations and the model's adaptability in complex geographical environments, providing effective strategies and insights for future geographical named entity recommendations in similar scenarios. Comparing the experimental results from Figure 8d–f, it is evident that incorporating RAG technology can effectively enhance the accuracy and relevance of recommendations in complex geographical named entity matching tasks, particularly when addressing detailed matching requirements. In addressing recommendation tasks involving specific geographical named entity elements, combining matching models with RAG technology not only enhances the accuracy and relevance of address recommendations but also reveals potential data quality issues in address databases, thus offering valuable insights for the optimization of future address recommendation systems. Overall, RAG, by incorporating an external additional address knowledge base, provides a significant supplement to deep-learning-based matching models for geographical named entity recommendation tasks. The generative model, when handling externally redundant RAG retrieval information, can identify and extract key geographical named entity elements, thus achieving better matching and recommendations. Through entity recommendations in various case types, we have validated the effectiveness and feasibility of this method in geographical named entity recommendations.

5. Conclusions and Future Work

This study developed a multi-task oriented model for matching geographical named entities. Initially, we conducted combination experiments on the sub-task loss function weights of the model to determine the optimal weight combination. Subsequently, through ablation studies on various sub-tasks, we compared accuracy metrics across multiple task combinations, validating the effectiveness of the multi-task decoder architecture. Furthermore, we conducted comparative experiments with other models, showing that the multi-task training approach significantly enhances performance in geographical named entity matching tasks. We then undertook geographical named entity recommendation tasks, comparing our model's distance error in address matching with that of several traditional models, demonstrating our model's superiority in recommendation tasks. Finally, we executed geographical named entity recommendation tasks based on Retrieval-Augmented Generation (RAG), confirming the efficacy of our model in matching different types of geographical entities. By selecting various types for RAG-based recommendation experiments, the results indicate that RAG enhances geographical named entity recommendation outcomes, proving its effectiveness and feasibility.

Although this study has achieved certain results in the field of geographic named entity matching, there are some limitations. For example, the address dataset used for training and matching, while large-scale with millions of entries, may limit the performance of multi-task models. Additionally, the external address information introduced by RAG contains redundancies, and the generative model sometimes struggles to extract the relevant information needed for further recommendations. Currently, this technology is primarily applicable to Chinese addresses equipped with latitude and longitude information, which may limit its generalizability in handling addresses without coordinates or in non-Chinese

language environments. To overcome these limitations and expand the application scope of the technology stack, future work will involve introducing address data in multiple languages, such as English and Spanish, and using geocoding technology to estimate the latitude and longitude of addresses without coordinates. By testing and optimizing the model across different languages and cultural backgrounds, we aim to enhance the model's applicability and accuracy. We also plan to develop a more effective RAG module tailored to geographic named entities to better handle complex queries related to geographic locations. Further research will also consider how to better understand and utilize RAG technology to handle the complex relationships between geographic named entities, especially in varied and diverse urban environments.

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